

Into the Future

Far from seeing the aircraft market collapsing as fossil fuels continue to surge in price, there is a polarization of aircraft manufacture. At one end, the giant aircraft-building nations are pouring resources into state-of-the-art, next-generation fighters and light, efficient airliners, while at the other end of the scale, small manufacturers are popping up building ultralight electric-powered aircraft charged by solar or wind power.



△ White Knight One & SpaceShipOne 2003

Origin US

Engine White Knight: 2 x 2,400-3,600 lb (1,089-1633 kg) thrust General Electric J85-GE-5 afterburning turbojet

Top speed 445 mph (716 km/h)

Burt Rutan's creation achieved the first manned private spaceflight with SpaceShipOne for astronaut Mike Melville, after airborne launch from its mother ship, White Knight. The spacecraft flew at 2,292 mph (3,689 km/h), powered by an N2O/HTPB Spacedev hybrid rocket.

SpaceShipOne attached to main aircraft

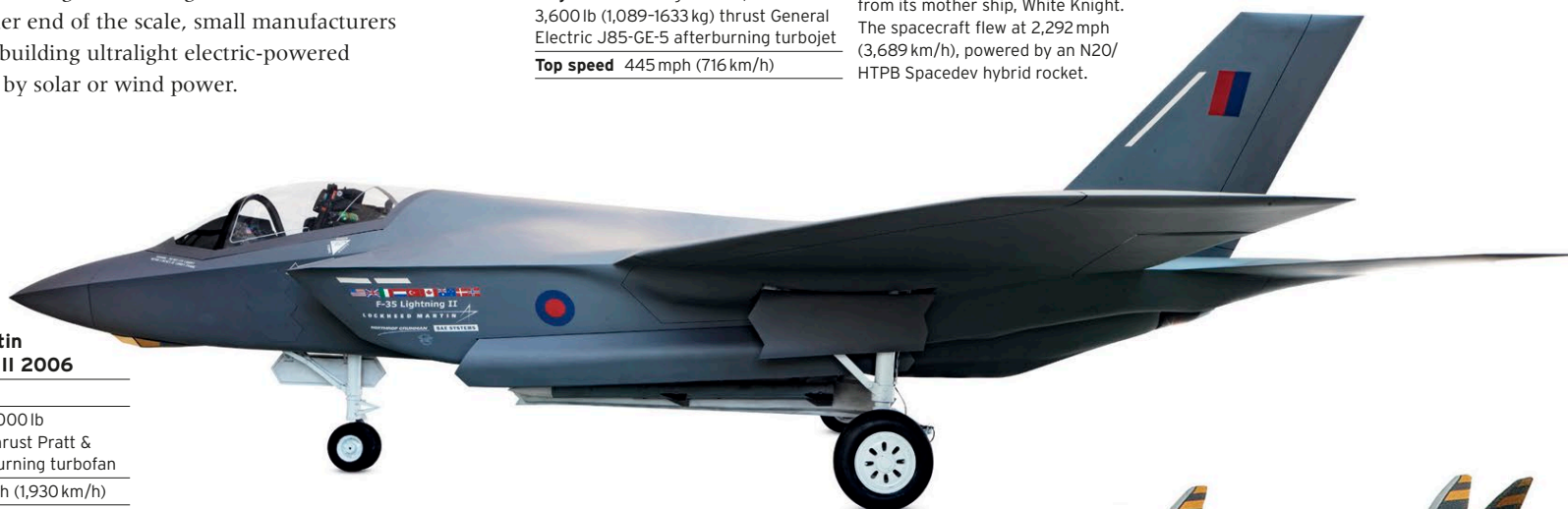
▷ Lockheed Martin F-35A Lightning II 2006

Origin US

Engine 28,000-43,000 lb (12,700-19,500 kg) thrust Pratt & Whitney F135 afterburning turbofan

Top speed 1,200 mph (1,930 km/h)

The advanced-stealth-technology F-35 has three forms: F-35A conventional takeoff and landing; F-35B short takeoff, vertical landing; F-35C carrier-based. It will equip most of the western world.



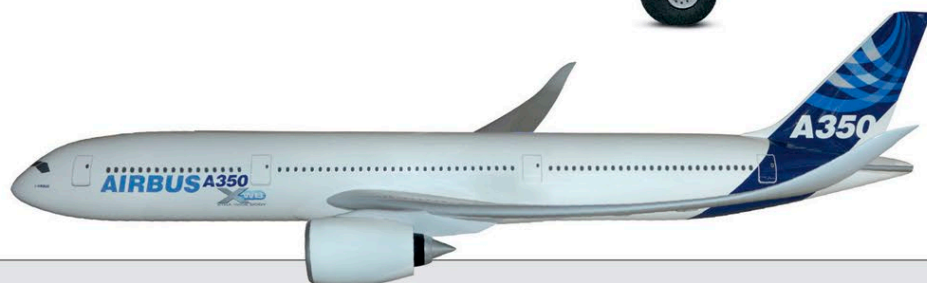
▷ Airbus A400M Atlas 2009

Origin European consortium

Engine 4 x 11,060 hp Europrop TP400-D6 turboprop

Top speed 485 mph (780 km/h)

Immersed in controversy over delays and overspending, the all-new A400M Atlas doubles the military long-range transport capacity of the C-130 Hercules and boasts state-of-the-art technology.



◁ Airbus A350-800 2014

Origin European consortium

Engine 2 x 84,000 lb (38,102 kg) Rolls-Royce Trent XWB-83 turbofan

Top speed 587 mph (945 km/h)

Customer pressure led Airbus to a radical overhaul for its new family of wide-body jets. They used carbon fiber-reinforced polymer for both fuselage and wings as part of a major efficiency drive.

Helicopter Advances

While cost and relatively high fuel consumption has prevented helicopters becoming mainstream aircraft, their unique vertical takeoff and landing capabilities ensure their popularity. Composite construction has helped reduce weight, while experiments continue with booster engines and propellers to improve maximum speeds and unmanned models controlled by computers.



▷ Eurocopter X3 2010

Origin France

Engine 2 x 2270 shp Turbomeca RTM332 turboshaft

Top speed 267 mph (430 km/h)

As a less-complex solution to the problems of high-speed helicopter flight than a tilt-rotor, the X3's rotor can be slowed down to keep tip speeds out of the efficiency-sapping transonic range.